Birth Order and Academic Productivity in Pre-Modern Europe

Evidence from Brothers in Universities and Learned Academies

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Remote dissertation defense

Research question and contribution

Did older brothers among pre-modern European scholars become more productive than their younger brothers?

Why this is interesting

- Modern evidence often finds firstborn advantages in education and long-run outcomes.
- Historical evidence is more mixed and institutional context differs sharply from today.
- Scholarly families offer a setting in which long-run intellectual output can be observed.

What this dissertation adds

- A new sample of **645 rankable brother pairs**.
- Direct **within-family older – younger comparisons**.
- A distinction between **entry into publication** and productivity conditional on publishing.
- Links to **academic institutions** and **regional inheritance regimes**.

Literature and hypotheses

1. Birth order and human capital

Modern within-family studies generally find an earlier-born advantage.

Devereux et al. (2005); Behrman & Taubman (1986); Kantarevic & Mechoulan (2006)

2. Mechanisms

Resource dilution, differential parental investment, and sibling competition.

Becker & Tomes (1976); Blake (1981); Price (2008); Hotz & Pantano (2015)

3. Historical and elite settings

Historical effects are smaller or more context-dependent; elite formation and institutions matter.

Clark & Cummins (2024); Cools et al. (2024); De la Croix & Goñi (2024)

4. Family institutions

Inheritance systems can alter the allocation of resources within families.

Todd (1990); Gay et al. (2024); Gobbi et al. (2026)

Ex ante hypothesis: older-brother advantages should be larger where family resources are less equally distributed; institutional environments may also amplify sibling divergence.

Data and sample construction

Sources

- **RETE (Repertorium Eruditorum Totius Europae)**
  Scholars in European universities and learned academies; birth information and academic careers.
- **Biographical repertories**
  Historical catalogues of professors and academic elites; additional family links (fathers, sons and brothers).
- **VIAF (Virtual International Authority File)**
  Bibliographic productivity: titles, publishers, publication countries and coauthors.
- **Wikipedia / Wikidata**
  Scholarly visibility (page length and languages) and academic affiliations / occupations.
- **Todd (1990) + Gay et al. (2024)**
  Regional inheritance systems.

Sample construction

913	645	268
identified brother pairs	rankable pairs	unrankable pairs

Rankable = usable for birth-order analysis

Both birth years are observed and different, so older and younger brothers can be assigned without arbitrary choices.

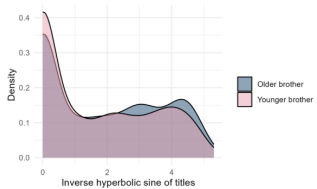
Selection

Rankable pairs are more visible in the historical record than unrankable pairs. Results therefore describe **observed scholarly elites**, not all families.

Outcome and transformation

Main outcome

- Number of titles attributed to each scholar in VIAF.
- Alternative outcomes capture visibility, publishers, coauthors, languages, and institutional involvement.



Why transform? Titles are strongly right-skewed, with many observations close to zero.

Inverse hyperbolic sine transformation

$$\operatorname{asinh}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$$

Key property

Defined at zero, while behaving approximately like a logarithm for large positive values.

Interpretation

Asinh coefficients are **not constant percentage effects near zero**. Percentage interpretations depend on the chosen baseline.

Descriptive within-family evidence

Paired comparison

For each brother pair:

$$d_i = Outcome_i^{Older} - Outcome_i^{Younger}$$

$$H_0 : E[d_i] = 0$$

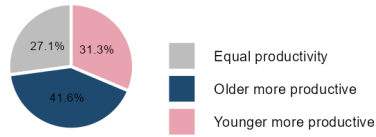
$E[d_i] > 0 \Rightarrow$ older brothers have higher values on average.

Measure	Mean diff.	p-val.
Titles	2.096	.0076
Titles asinh	0.245	.0007
Has title	0.056	.0089
Wiki languages	0.659	.0215
Publishers	0.667	.0043
Coauthors	1.056	.0001

Takeaway

Average differences generally favor older brothers, but the advantage is **modest and far from universal**.

Who is more productive?



Heterogeneity across families

Older brother	**41.6%**
Younger brother	**31.3%**
Equal productivity	**27.1%**

The average advantage favors older brothers, but nearly **6 in 10 pairs** do not show an older-brother advantage.

Where does the difference come from? Extensive vs. intensive margin

Extensive margin: any recorded title

$$Pr(Title_i > 0) = \Lambda(\alpha + \beta Older_i + \gamma AgeGap_i + \delta FatherAcademia_i)$$

Logit result

Older-sibling coefficient: **0.238*****
Odds ratio: $e^{0.238} = \mathbf{1.27}$

Interpretation: older brothers have about **27% higher odds** of having at least one recorded title, conditional on included controls.

$N = 1{,}290$ scholar observations $= 645$ pairs $\times 2$ brothers.

All scholars remain in the sample because the outcome is whether $Title > 0$.

Intensive margin: output among publishers

$$asinh(Titles_i) = \alpha + \beta Older_i + \gamma AgeGap_i + \delta FatherAcademia_i + \varepsilon_i$$

OLS on positive-title sample

Older-sibling coefficient: **0.141**
Not statistically significant.

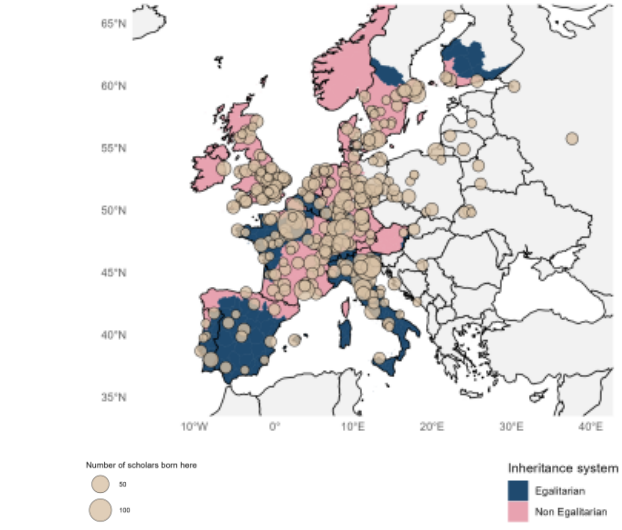
$N = 772$ scholars with positive title counts.

Scholars with zero titles are excluded because this specification measures productivity **conditional on publishing**.

Interpretation

The strongest birth-order pattern concerns **participation in recorded scholarly production**, not output once a scholar is already publishing.

Institutional environment and inheritance systems



Inheritance

- Assigned by birthplace region at NUTS3 level.
- Egalitarian: **30.1%**
- Non-egalitarian: **46.5%**
- Unknown: **23.4%**

Institutions

- Indicator $= 1$ if brothers are associated with different academic institutions.

Important

"Older observed scholar" is not necessarily the biological firstborn. The inheritance variable is a **regional proxy**, not a household rule.

Empirical specification

$$\Delta P_i = \text{asinh}(\mathit{Titles}_i^{\mathit{Older}}) - \text{asinh}(\mathit{Titles}_i^{\mathit{Younger}})$$

$$\Delta P_i = \beta_0 + \beta_1 \mathit{AgeGap}_i + \beta_2 \mathit{FatherAcademia}_i + \beta_3 \mathit{DifferentInstitution}_i$$
$$+ \beta_4 \mathit{NonEgalitarian}_i + \beta_5 \mathit{Unknown}_i + \sum_{c=13}^{17} \gamma_c \mathit{Century}_{ic} + \varepsilon_i$$

- Positive ΔP_i: higher recorded productivity for the older brother.
- OLS with standard errors clustered at family level; 18th century omitted.
- Pair differencing removes shared family-level levels, but does not randomize birth order, survival, observation, career choice or institution.

Identification claim

I interpret these as **within-family associations**, not fully causal estimates.

Preferred regression: institutions are the most robust correlate

Variable	Coefficient	S.E.	Significance
Age gap	0.019	0.011	*
Father in academia	-0.163	0.150	n.s.
Different institution	**0.485**	0.180	***
Non-egalitarian inheritance	0.135	0.175	n.s.
Unknown inheritance	0.130	0.205	n.s.

What I would emphasize

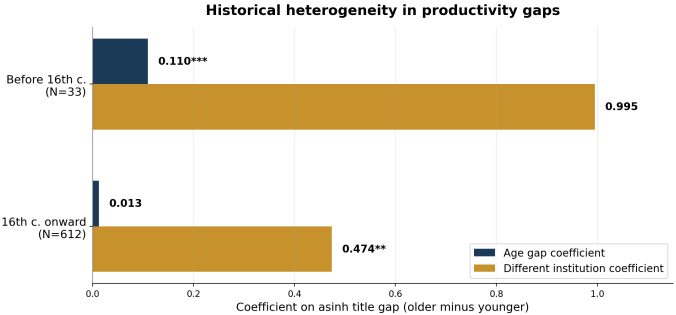
The coefficient on **different institution** is positive and statistically robust: brothers associated with different institutions display larger older – younger productivity gaps.

What I would not claim

The coefficient does not establish that moving brothers to different institutions *causes* a 63% increase in productivity. It is an association in transformed gap units.

Preferred model: $N = 645$, $R^2 = 0.035$, century indicators included, family-clustered SE.

Historical heterogeneity



Period comparison

Before the 16th century

- **33 pairs**
- Age gap: **0.110*****
- Different institution: 0.995, imprecise

16th century onward

- **612 pairs**
- Age gap: 0.013, n.s.
- Different institution: **0.474****

Why split at the 16th century?

Inheritance-system variation is very limited in the earliest centuries: **non-egalitarian regimes are almost absent in the 13th and 14th centuries**. The later period therefore provides substantially more variation in inheritance regimes.

Caution

The early-period estimates are based on only **33 brother pairs**.
They are informative, but substantially less precise and should therefore be interpreted cautiously.

Robustness and main threats to interpretation

Robustness: what does each test check?

- **Age gap ≤ 20 years**
  Excludes unusually large sibling age gaps.
- **Two observed sons**
  Avoids multiple overlapping brother pairs in larger families.
- **Alternative outcomes**
  Checks that results are not specific to VIAF titles.
- **Alternative functional forms**
  Tests dependence on the asinh transformation.
- **Country controls**
  Checks whether geographic composition drives the result.

Stable pattern

Institutional divergence coefficients remain positive; inheritance coefficients remain weak.

Main threats / limitations

- **Historical selection**
  Results apply to observed scholarly elites.
- **Rankability selection**
  Better-documented scholars are overrepresented.
- **VIAF measurement**
  Titles also reflect preservation, editions and cataloguing.
- **Endogenous institutions**
  Institution choice may reflect ability or opportunities.
- **Inheritance proxy**
  Regional regimes may differ from household practice.

Conclusion

1. There is a **modest older-brother advantage** in recorded scholarly productivity.
2. The clearest birth-order pattern is on the **extensive margin**: older brothers have higher odds of having at least one recorded title.
3. Conditional on publishing, the older-sibling coefficient is small and statistically insignificant.
4. **Institutional divergence** is more consistently associated with within-family productivity gaps than regional inheritance regimes.
5. The relationship is historically heterogeneous: age gaps matter more in the very small early sample; institutional divergence is more robust from the 16th century onward.

The evidence points less to a universal primogeniture effect than to birth-order differences shaped by access to scholarly activity and institutional context.

Thank you

Questions and discussion

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Backup: full preferred regression

Variable	Coefficient	S.E.
Constant	-0.162	0.197
Age gap	0.019*	0.011
Father in academia	-0.163	0.150
Different institution	0.485***	0.180
Non-egalitarian inheritance	0.135	0.175
Unknown inheritance	0.130	0.205
13th century	0.099	0.223
14th century	1.764***	0.552
15th century	0.863**	0.345
16th century	0.025	0.226
17th century	0.099	0.161
Observations	645	
R^2	0.035	
Adj. R^2	0.020	

Dependent variable: $\text{asinh}(Titles_O) - \text{asinh}(Titles_Y)$. 18th century omitted. Family-clustered SE.

Backup: sample selection

	Rankable	Unrankable
Brother pairs	645	268
Mean titles	10.84	1.87
Individuals with no titles	40.2%	81.6%
Pairs with no titles	25.4%	69.3%
Father in academia	38.8%	41.0%

- Birth order is observable disproportionately for better documented scholars.
- This can affect external validity.
- It is not solved by the within-family comparison.

Backup: concrete examples from the data

Bernoulli brothers: similar productivity

Jacob Bernoulli
Older brother: **63 VIAF titles**

Johann Bernoulli
Younger brother: **59 VIAF titles**

A case where birth order is associated with only a very small within-family productivity difference.

Thomasius brothers: large productivity gap

Christian Thomasius
Older brother: **71 VIAF titles**

Gottfried Thomasius
Younger brother: **5 VIAF titles**

One of the largest within-family productivity gaps observed in the dataset.

Schott brothers: low visibility, but observable in structured data

August Ludwig Schott (1751–1787): 4 titles **Andreas Heinrich Schott (1758–1831): 6 titles**

Neither brother has a significant presence in accessible historical sources. The pair is identified through **prosopographical data**, illustrating how structured historical datasets recover scholars who would otherwise be difficult to observe.

Takeaway: the dataset contains both highly visible scholarly families and much less visible pairs, with very different within-family productivity patterns.

Backup: Scholarly Productivity Across Samples

Sample	Mean Titles	Observations
All scholars in the dataset	3.81	91,931
Scholars with identified siblings	8.29	1,552
Older brothers (rankable)	11.89	645
Younger brothers (rankable)	9.79	645

Key Takeaways

- Scholars with identified siblings have higher recorded productivity than the overall sample.
- Older brothers have, on average, 2.10 more recorded titles than younger brothers.
- The median number of titles in the full dataset is zero.

Notes: The full historical academic dataset includes 91,931 scholars. Among the 1,556 unique scholars with identified siblings, title data are available for 1,552 (four have missing values). Zero-title observations are included in the means. Older and younger brother means are based on 645 rankable pairs rather than unique individuals.

Backup: asinh interpretation

Transformations

$$\operatorname{asinh}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$$

$$\sinh(z) = \frac{e^z - e^{-z}}{2}$$

Percentage illustration

$$\%\Delta y = 100 \left[\frac{\sinh(\operatorname{asinh}(y_0) + \beta)}{y_0} - 1 \right]$$

- This is a **baseline-dependent illustration**, not a constant semi-elasticity.
- Using $y_0 = 9.79$:
  - $\beta = 0.245 \Rightarrow \%\Delta y \approx$ **27.9%**
  - $\beta = 0.485 \Rightarrow \%\Delta y \approx$ **62.7%**

Interpretation

The percentage changes depend on the chosen baseline y_0. For this reason, the main results should be reported first in **asinh units**, with percentage conversions used only as illustrations.

Backup: extensive and intensive margins

	Extensive (Logit) $Pr(Title > 0)$	Intensive (OLS) Asinh titles
Constant	0.051	2.892***
Older sibling	**0.238*** **	0.141
Age gap	-0.002	-0.013
Father in academia	0.654***	0.090
Observations	1290	772
Family-clustered SE	Yes	Yes

Logit interpretation

$$OR = e^{\beta} = e^{0.238} \approx 1.27$$

$$(1.27 - 1) \times 100 \approx \textbf{27\% higher odds}$$

So older brothers have about **27% higher odds** of having at least one recorded title, conditional on the included controls.

Important

This is an increase in **odds**, not a 27% increase in probability.

Backup: period-specific regression

	Before 16th c.	16th c. onward
Observations	33	612
Age gap	0.110***	0.013
Father in academia	0.415	-0.160
Different institution	0.995	0.474**
Non-egalitarian inheritance	-0.719	0.164
Unknown inheritance	-1.730	0.175
R^2	0.387	0.020
Adj. R^2	0.215	0.008

Backup: robustness across functional forms

Specification	Different institution	Main caution
Asinh gap	0.485***	Baseline-dependent percentage translation
Log gap, $\epsilon = 0.1$	0.577**	Depends on zero adjustment
Raw title gap	5.160**	More sensitive to outliers
Age gap ≤ 20	0.478***	$N = 624$
Two-son families	0.493**	$N = 467$
Country controls	0.388*	Many countries have small cell sizes

Takeaway

The sign and broad qualitative pattern for institutional divergence survive several alternative specifications, although precision and magnitude vary.

Backup: literature map

- **Modern birth order:** Devereux et al. (2005), Behrman & Taubman (1986), Kantarevic & Mechoulan (2006). Earlier-born children tend to have better schooling outcomes in modern within-family data.
- **Mechanisms:** Becker & Tomes (1976), Blake (1981), Price (2008), Hotz & Pantano (2015). Finite resources, parental time, strategic responses and sibling competition.
- **Historical evidence:** Clark & Cummins (2024), Cools et al. (2024), Nuevo-Chiquero et al. (2022). Historical birth-order effects are mixed and context-dependent.
- **Academic elites:** De la Croix & Goñi (2024), De la Croix et al. (2024). Family background, institutions and networks matter in historical academia.
- **Inheritance institutions:** Todd (1990), Gay et al. (2024), Gobbi et al. (2026). Regional family systems vary spatially and can affect household incentives.

Backup: likely econometrics questions

- **Causal?** Within-family differencing removes shared family levels, but not selection, sibling-specific shocks, endogenous institution choice or measurement error.
- **Why cluster by family?** Larger families generate several pair observations; family clustering allows residual correlation within reconstructed families.
- **Why not count models?** The estimate is an older–younger within-pair gap including zeros; asinh gives a symmetric transformed difference. Count models are a useful extension, not the same estimand.
- **Why not pair fixed effects?** The regression is already at pair level; the dependent variable is the within-pair difference, so shared pair-level levels cancel.
- **Low R^2?** The goal is not prediction; historical productivity is highly heterogeneous. The question is whether selected covariates are systematically associated with the gap.

Backup: questions about measurement and selection

- **Does VIAF measure true productivity?** No. It is a bibliographic proxy for recorded output and also reflects editions, translations, preservation and cataloguing.
- **Why compare scholars only?** The research question concerns observed academic elites; it does not estimate birth-order effects for the general population.
- **Is older the same as firstborn?** No. I identify relative age among observed scholarly brothers, not complete biological birth rank.
- **Could different institutions reflect ability?** Yes. Institution choice is endogenous; I interpret it as an association with divergent trajectories, not a causal institution effect.
- **Could rankability bias results?** Yes. Rankable pairs have richer records, so the external validity is selected and should be stated explicitly.